

Docker 기본명령어

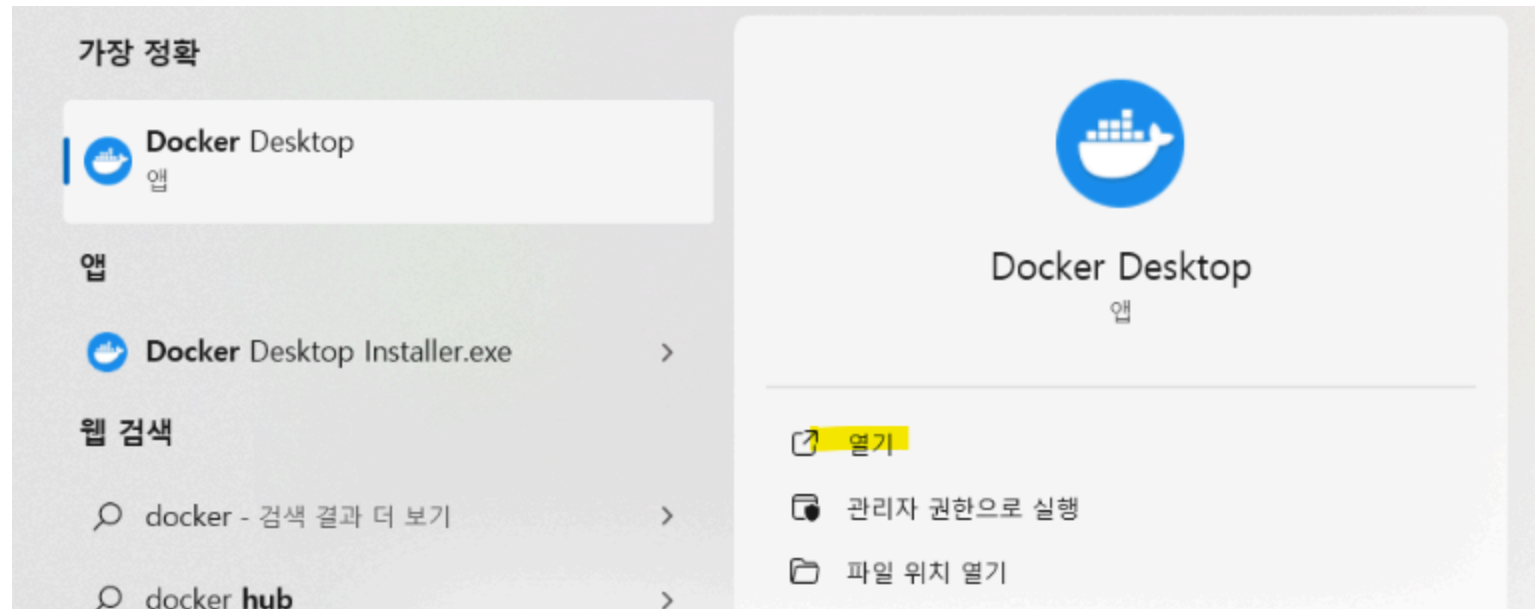
참고문서

- <https://cultivo-hy.github.io/docker/image/usage/2019/03/14/Docker정리/>
- <https://pyrasis.com/Docker/Docker-HOWTO>

자주 사용하는 명령어 옵션들

옵션	설명
-d	detached mode 흔히 말하는 백그라운드 모드
-p	호스트와 컨테이너의 포트를 연결 (포워딩)
-v	호스트와 컨테이너의 디렉토리를 연결 (마운트)
-e	컨테이너 내에서 사용할 환경변수 설정
-name	컨테이너 이름 설정
-rm	프로세스 종료시 컨테이너 자동 제거
-it	-i와 -t를 동시에 사용한 것으로 터미널 입력을 위한 옵션
-link	컨테이너 연결 [컨테이너명:별칭]

Docker Desktop 실행(열기)



search

- `docker search <이미지명>` : [Docker Hub](#)에서 해당 이미지들을 검색할 수 있습니다.

```
Windows PowerShell
새로운 기능 및 개선 사항에 대한 최신 PowerShell을 설치하세요! https://aka.ms/PSWindows

PS C:\Users\Playdataa> docker search ubuntu
```

NAME	DESCRIPTION	STARS	OFFICIAL	AUTOMATED
ubuntu	Ubuntu is a Debian-based Linux operating sys...	16458	[OK]	
websphere-liberty	WebSphere Liberty multi-architecture images ...	297	[OK]	
open-liberty	Open Liberty multi-architecture images based...	62	[OK]	
neurodebian	NeuroDebian provides neuroscience research s...	104	[OK]	
ubuntu-debootstrap	DEPRECATED; use "ubuntu" instead	52	[OK]	
ubuntu-upstart	DEPRECATED, as is Upstart (find other proces...	115	[OK]	
ubuntu/nginx	Nginx, a high-performance reverse proxy & we...	100		
ubuntu/squid	Squid is a caching proxy for the Web. Long-t...	67		
ubuntu/cortex	Cortex provides storage for Prometheus. Long...	4		
ubuntu/apache2	Apache, a secure & extensible open-source HT...	60		

pull

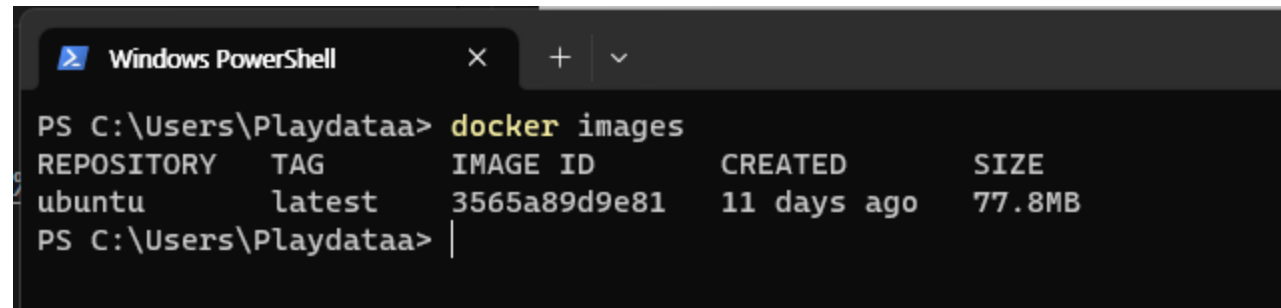
- `docker pull <이미지명>:<태그>` : 이미지를 받아올 수 있습니다.

```
Windows PowerShell
PS C:\Users\Playdataa> docker pull ubuntu:latest
latest: Pulling from library/ubuntu
37aaf24cf781: Pull complete
Digest: sha256:9b8dec3bf938bc80fbe758d856e96fdfab5f56c39d44b0cff351e847bb1b01ea
Status: Downloaded newer image for ubuntu:latest
docker.io/library/ubuntu:latest

What's Next?
  View a summary of image vulnerabilities and recommendations → docker scout quickview ubuntu:latest
PS C:\Users\Playdataa> |
```

images

- `docker images` : 다운로드 받은 이미지의 목록을 출력할 수 있습니다.



```
Windows PowerShell
PS C:\Users\Playdataa> docker images
REPOSITORY    TAG       IMAGE ID      CREATED       SIZE
ubuntu        latest    3565a89d9e81  11 days ago  77.8MB
PS C:\Users\Playdataa> |
```

-  Containers
-  Images
-  Volumes
-  Dev Environments BETA
-  Docker Scout
-  Learning center

Extensions 

 Add Extensions

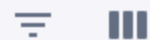
Images [Give feedback](#)

Local Hub Artifactory EARLY ACCESS

77.82 MB / 0 Bytes in use 1 images

Last refresh: 23 hours ago 

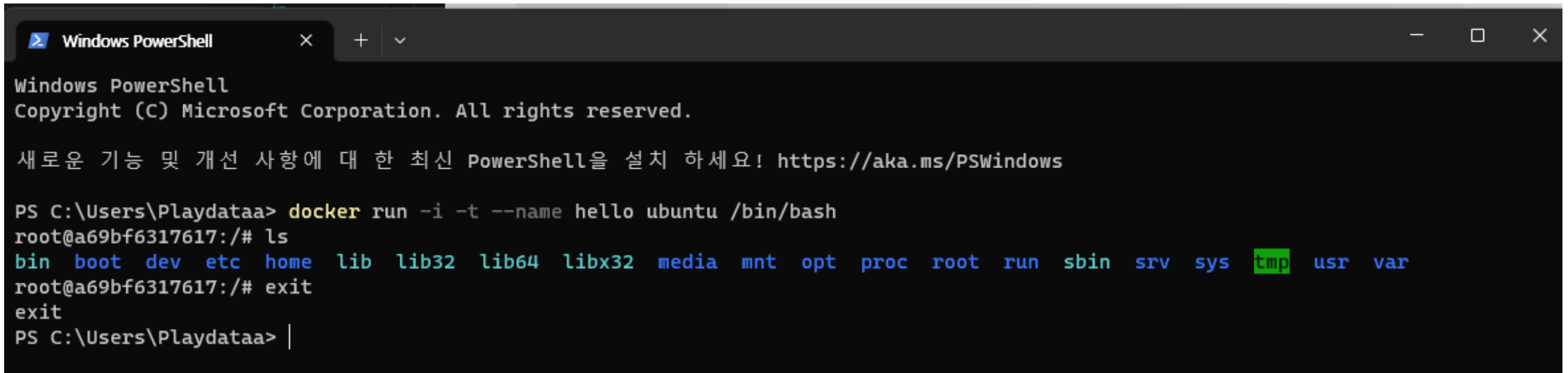
 Search



☐	Name	Tag	Status	Created	Size	Actions
☐	ubuntu 3565a89d9e81 	latest	In use	11 days ago	77.82 MB	  

run

- `docker run -i -t --name <컨테이너이름> ubuntu /bin/bash`
 - `-i` : 보통 이 옵션을 사용하여 Bash 에 명령을 입력합니다.
 - `-t` : Bash를 사용하려면 이 옵션을 설정해야 합니다.
 - `--name` : 컨테이너 이름을 설정합니다.



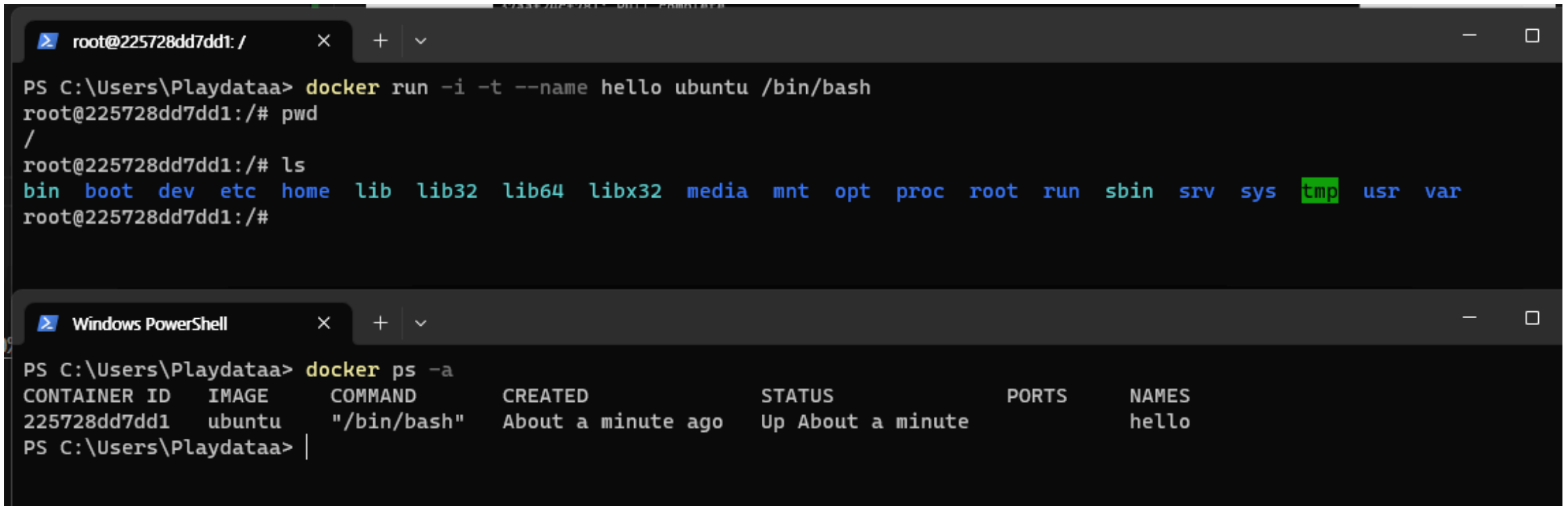
```
Windows PowerShell
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PS C:\Users\Playdataa> docker run -i -t --name hello ubuntu /bin/bash
root@a69bf6317617:/# ls
bin boot dev etc home lib lib32 lib64 libx32 media mnt opt proc root run sbin srv sys tmp usr var
root@a69bf6317617:/# exit
exit
PS C:\Users\Playdataa> |
```

ps

- `docker ps -a` : 모든 컨테이너 목록을 출력합니다.
 - `-a` : 사용이 정지된 컨테이너까지 모두 출력



```
PS C:\Users\Playdataa> docker run -i -t --name hello ubuntu /bin/bash
root@225728dd7dd1:/# pwd
/
root@225728dd7dd1:/# ls
bin boot dev etc home lib lib32 lib64 libx32 media mnt opt proc root run sbin srv sys tmp usr var
root@225728dd7dd1:/#
```

```
PS C:\Users\Playdataa> docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS   NAMES
225728dd7dd1   ubuntu    "/bin/bash"             About a minute ago Up About a minute           hello
PS C:\Users\Playdataa> |
```

stop

- `docker stop <컨테이너 이름>` : 컨테이너 멈추기

```
Windows PowerShell
PS C:\Users\Playdataa> docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS   NAMES
225728dd7dd1   ubuntu   "/bin/bash"   7 minutes ago   Up 7 minutes           hello
PS C:\Users\Playdataa> docker stop hello
hello
PS C:\Users\Playdataa> docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS   NAMES
225728dd7dd1   ubuntu   "/bin/bash"   8 minutes ago   Exited (137) 6 seconds ago           hello
PS C:\Users\Playdataa> |
```

start

- `docker start <컨테이너 이름>` : 컨테이너 실행하기

```
Windows PowerShell
PS C:\Users\Playdataa> docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS              PORTS          NAMES
a69bf6317617   ubuntu   "/bin/bash"             4 minutes ago Exited (0) 3 minutes ago
PS C:\Users\Playdataa> docker start hello
hello
PS C:\Users\Playdataa> docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS              PORTS          NAMES
a69bf6317617   ubuntu   "/bin/bash"             4 minutes ago Up 4 seconds      hello
PS C:\Users\Playdataa> |
```

restart

- `docker restart <컨테이너 이름>` : 컨테이너 재실행하기

```
Windows PowerShell
PS C:\Users\Playdataa> docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS   NAMES
a69bf6317617   ubuntu   "/bin/bash"             8 minutes ago Up 4 minutes             hello
PS C:\Users\Playdataa> docker restart hello
hello
PS C:\Users\Playdataa> docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS   NAMES
a69bf6317617   ubuntu   "/bin/bash"             9 minutes ago Up 3 seconds             hello
PS C:\Users\Playdataa> |
```

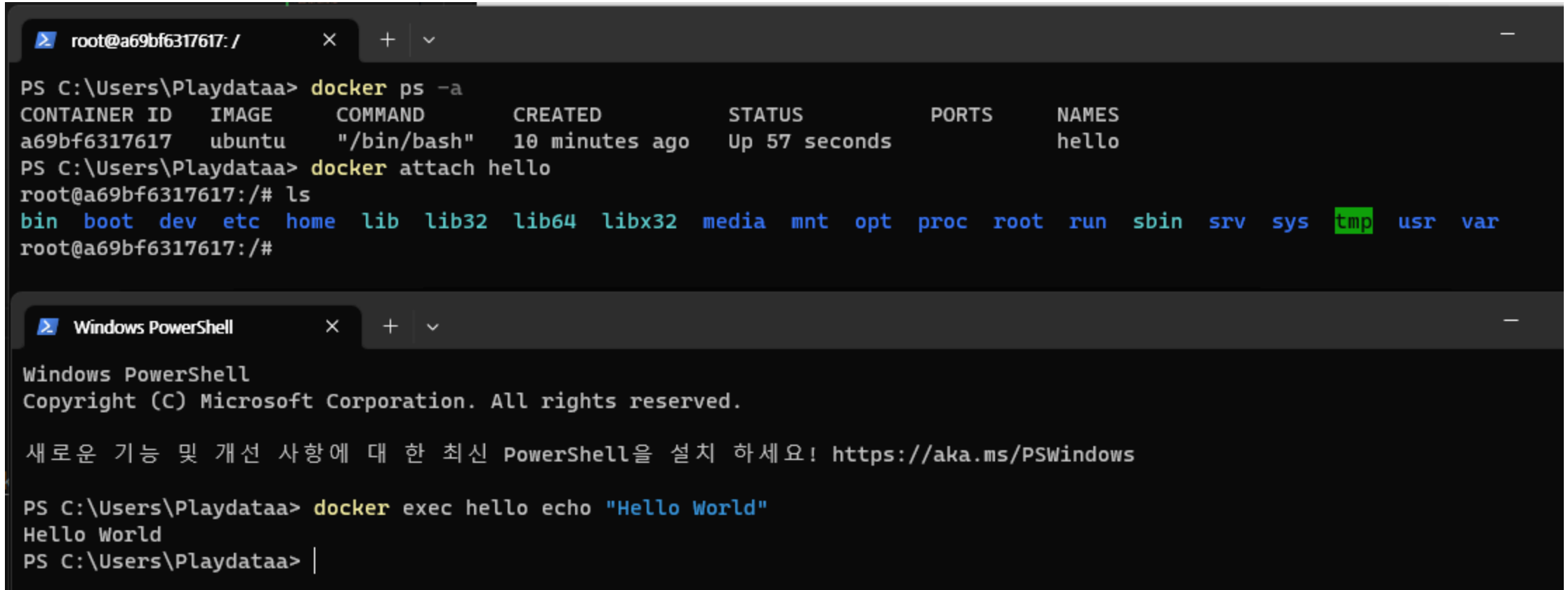
attach

- `docker attach hello` : 컨테이너에 접속하기

```
root@28d00d39d17e: /
PS C:\Users\Playdataa> docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS   NAMES
28d00d39d17e   ubuntu   "/bin/bash"             36 seconds ago Up 4 seconds   hello
PS C:\Users\Playdataa> docker attach hello
root@28d00d39d17e:/# ps
  PID TTY          TIME CMD
    1 pts/0        00:00:00 bash
    9 pts/0        00:00:00 ps
root@28d00d39d17e:/# ls
bin  boot  dev  etc  home  lib  lib32  lib64  libx32  media  mnt  opt  proc  root  run  sbin  srv  sys  tmp  usr  var
root@28d00d39d17e:/# |
```

exec

- `docker exec <컨테이너이름> echo "hello World"` : 외부에서 컨테이너에 명령하기



```
root@a69bf6317617: /
PS C:\Users\Playdataa> docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS   NAMES
a69bf6317617   ubuntu   "/bin/bash"             10 minutes ago Up 57 seconds        hello
PS C:\Users\Playdataa> docker attach hello
root@a69bf6317617:/# ls
bin boot dev etc home lib lib32 lib64 libx32 media mnt opt proc root run sbin srv sys tmp usr var
root@a69bf6317617:/#

Windows PowerShell
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PS C:\Users\Playdataa> docker exec hello echo "Hello World"
Hello World
PS C:\Users\Playdataa> |
```

rm

- `docker rm <컨테이너이름>` : 컨테이너 삭제

```
Windows PowerShell
PS C:\Users\Playdataa> docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS        NAMES
a69bf6317617   ubuntu   "/bin/bash"             13 minutes ago Exited (0) 6 seconds ago
PS C:\Users\Playdataa> docker rm hello
hello
PS C:\Users\Playdataa> docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS        NAMES
PS C:\Users\Playdataa> |
```

rmi

- `docker rmi <이미지명>:<태그>` : 이미지 삭제

```
Windows PowerShell
PS C:\Users\Playdataa> docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
ubuntu        latest    3565a89d9e81   11 days ago    77.8MB
PS C:\Users\Playdataa> docker rmi ubuntu:latest
Untagged: ubuntu:latest
Untagged: ubuntu@sha256:9b8dec3bf938bc80fbe758d856e96fdfab5f56c39d44b0cff351e847bb1b01ea
Deleted: sha256:3565a89d9e81a4cb4cb2b0d947c7c11227a3f358dc216d19fc54bfd77cd5b542
Deleted: sha256:01d4e4b4f381ac5a9964a14a650d7c074a2aa6e0789985d843f8eb3070b58f7d
PS C:\Users\Playdataa> docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
PS C:\Users\Playdataa> |
```